

Detailed ORBIT Equivalence Audit Through Mass Dimension 6

ORBIT

2026-03-23 20:38:09

Contents

1 Claim Being Verified

This report audits the statement that the canonically normalized Matchete-derived EFT matches the sign-flipped Einstein-Hilbert reference through mass dimension six after imposing the verified matching branch, reducing modulo IBP, reducing modulo the interaction-sector Fierz–Pauli field-redefinition image, and completing the quadratic four-derivative sector by quadratic higher-derivative field redefinitions.

The stored final summary used for this report is:

```
<|"TimingSeconds" -> <|"InputTruncation" -> 4.544105, "ReferenceExpansion" -> 0.005974, "Input"
```

2 Stored Inputs to the Audit

Top-level stored data used in the audit:

```
<|"ReferenceData" -> <|"MaxDimension" -> 6, "LambdaReferenceValue" -> LambdaRefFinal, "KappaRe"
```

3 Implemented Comparison Pipeline

This section mirrors the code path actually used in the stored run. Every major step below is quoted from the implementation files rather than reconstructed from memory.

3.1 Canonical normalization extraction

```
"219: kineticNormalizationDataFromReduction[inputReduction_Association, basisOpts_List] := Modu
```

3.2 Raw sector comparison records

```
"412: sectorComparisonRecord[key_, inputReduction_, referenceReduction_, sectorData_] := Modul
```

3.3 Matching-equation extraction

```
"433: collectDifferenceEquationRecords[sectorComparisons_Association] := Module[\n434:   {lmu =
```

3.4 Final verified exact-match logic

```
"581: finalizeComparisonResult[result_Association, basisOpts_List, opts_List] := Module[\n582:
```

3.5 Quadratic completion logic

```
"925: AnalyzeQuadraticFourDerivativeDifference[result_Association, opts : OptionsPattern[]] :=
```

3.6 Underlying IBP projection and interaction-sector redefinition image

```
"503: ProjectModuloIBP[expr_, basis_List] := Module[{ans, cleanExpr, eq, sol, qvars, coeffs},\n
```

```
"538: RedefinitionImageData[nh_Integer, Nd_Integer, ibpBasis_List, opts : OptionsPattern[RunBa
```

3.7 Independent cross-check script

```
"47: result = Get[resultFile];\n48: sec = ComputeSectorData[\n49: 2,\n50: 4,\n51:  \nMaxD
```

4 Canonical Normalization Check

The code compares the reduced $\{2, 2\}$ input coordinates to the reduced Fierz–Pauli coordinates and extracts a common factor Z_h .

Extracted canonical normalization data:

```
<|"SucceededQ" -> True, "Reason" -> "OK", "Factor" -> (M^2*\[Kappa]^2*(1 + Log[\[Mu]bar2/M^2]))
```

The resulting factor is:

```
(M^2*\[Kappa]^2*(1 + Log[\[Mu]bar2/M^2)))/24
```

The target kinetic sign used in the final comparison is -1.

5 Verified Branch Used in the Final Exact-Match Statement

The exact-match verdict does not come from a generic all-parameter symbolic solve. It comes from the explicit verification substitutions stored in the result object:

```
{Scalar[M] -> M, Scalar[\[Kappa]] -> \[Kappa], Scalar[\[Mu]bar2] -> \[Mu]bar2, Sqrt[1/(M^2*\[K
```

with assumptions

```
M > 0 && \[Kappa] > 0 && \[Mu]bar2 > 0
```

The code-level verified supported-sector check returned:

```
True
```

6 Supported-Sector Audit

For every supported sector, the report now lists the raw difference expression stored in the final result and the same expression after the verified substitutions used by `finalizeComparisonResult`.

Sector 1, 0

Raw difference expression:

$$(h[a, -a]*(-4*KappaRefFinal*LambdaRefFinal + 3*Sqrt[6]*M^4*\[Kappa]*Sqrt[-(1/((1 + Log[Scalar[$$

Difference after the verified substitutions used by `finalizeComparisonResult`:

$$0$$

Verified zero check: True.

Sector 2, 0

Raw difference expression:

$$(h[-a, -b]*h[a, b]*(KappaRefFinal^2*LambdaRefFinal + 9*M^2 + (KappaRefFinal^2*LambdaRefFinal +$$

Difference after the verified substitutions used by `finalizeComparisonResult`:

$$0$$

Verified zero check: True.

Sector 2, 2

Raw difference expression:

$$0$$

Difference after the verified substitutions used by `finalizeComparisonResult`:

$$0$$

Verified zero check: True.

Sector 3, 0

Raw difference expression:

$$(h[a, -a]*h[-b, -c]*h[b, c]*(KappaRefFinal^3*LambdaRefFinal + Log[Scalar[\[Mu]bar2]/Scalar[M]^2]$$

Difference after the verified substitutions used by `finalizeComparisonResult`:

$$0$$

Verified zero check: True.

Sector 3, 2

Raw difference expression:

$$(h[a, b]*(KappaRefFinal - 2*Sqrt[6]*\[Kappa]*Sqrt[-(1/((1 + Log[Scalar[\[Mu]bar2]/Scalar[M]^2]$$

Difference after the verified substitutions used by `finalizeComparisonResult`:

$$0$$

Verified zero check: True.

Sector 4, 0

Raw difference expression:

$$h[-a, c] * h[a, b] * h[-b, d] * h[-c, -d] * ((\text{KappaRefFinal}^4 * \text{LambdaRefFinal}) / 8 - 27 / (1 + \text{Log}[\text{Scalar}[\sqrt{\mu}]])$$

Difference after the verified substitutions used by `finalizeComparisonResult`:

$$0$$

Verified zero check: True.

Sector 4, 2

Raw difference expression:

$$h[-a, c] * h[a, b] * (-1/2 * \text{KappaRefFinal}^2 - 12 / ((1 + \text{Log}[\text{Scalar}[\sqrt{\mu}]]) / \text{Scalar}[M]^2)) * \text{Scalar}[M]$$

Difference after the verified substitutions used by `finalizeComparisonResult`:

$$0$$

Verified zero check: True.

Sector 5, 0

Raw difference expression:

$$-1/192 * (h[a, -a] * h[b, -b] * h[c, -c] * h[-d, -e] * h[d, e] * (7 * \text{KappaRefFinal}^5 * \text{LambdaRefFinal} * (1 + \text{Log}[\text{Scalar}[\sqrt{\mu}]])$$

Difference after the verified substitutions used by `finalizeComparisonResult`:

$$0$$

Verified zero check: True.

Sector 6, 0

Raw difference expression:

$$h[a, -a] * h[b, -b] * h[-c, e] * h[c, d] * h[-d, f] * h[-e, -f] * ((\text{KappaRefFinal}^6 * \text{LambdaRefFinal}) / 64 - ($$

Difference after the verified substitutions used by `finalizeComparisonResult`:

$$0$$

Verified zero check: True.

7 Exact Matching Equations Extracted by the Code

The function `collectDifferenceEquationRecords` expands each supported-sector coordinate difference, clears denominators with `Together`, and sets each coefficient of $L_\mu = \log(\bar{\mu}^2/M^2)$ to zero.

The resulting stored equations are:

Sector 1, 0, coordinate 1, polynomial equation 1:

$$-4*\text{KappaRefFinal}*\text{LambdaRefFinal} + 3*\text{Sqrt}[6]*M^4*\text{Kappa}*\text{Sqrt}[-(1/(M^2*\text{Kappa})^2*(1 + \text{Log}[\text{Mu}^2/M^2]))] == 0$$

Sector 2, 0, coordinate 1, polynomial equation 1:

$$\text{KappaRefFinal}^2*\text{LambdaRefFinal} + 9*M^2 + \text{KappaRefFinal}^2*\text{LambdaRefFinal}*\text{Log}[\text{Mu}^2/M^2] + 6*\text{KappaRefFinal}*\text{Sqrt}[6]*M^2 == 0$$

Sector 2, 0, coordinate 2, polynomial equation 1:

$$-(\text{KappaRefFinal}^2*\text{LambdaRefFinal}) - 9*M^2 - \text{KappaRefFinal}^2*\text{LambdaRefFinal}*\text{Log}[\text{Mu}^2/M^2] - 6*\text{KappaRefFinal}*\text{Sqrt}[6]*M^2 == 0$$

Sector 3, 0, coordinate 1, polynomial equation 1:

$$-(\text{KappaRefFinal}^3*\text{LambdaRefFinal}) - \text{KappaRefFinal}^3*\text{LambdaRefFinal}*\text{Log}[\text{Mu}^2/M^2] + 36*\text{KappaRefFinal}^2*\text{Sqrt}[6]*M^2 == 0$$

Sector 3, 0, coordinate 2, polynomial equation 1:

$$\text{KappaRefFinal}^3*\text{LambdaRefFinal} + \text{KappaRefFinal}^3*\text{LambdaRefFinal}*\text{Log}[\text{Mu}^2/M^2] - 36*\text{KappaRefFinal}^2*\text{Sqrt}[6]*M^2 == 0$$

Sector 3, 2, coordinate 1, polynomial equation 1:

$$\text{KappaRefFinal} - 2*\text{Sqrt}[6]*\text{Kappa}*\text{Sqrt}[-(1/(M^2*\text{Kappa})^2*(1 + \text{Log}[\text{Mu}^2/M^2])))] == 0$$

Sector 3, 2, coordinate 3, polynomial equation 1:

$$-\text{KappaRefFinal} + 2*\text{Sqrt}[6]*\text{Kappa}*\text{Sqrt}[-(1/(M^2*\text{Kappa})^2*(1 + \text{Log}[\text{Mu}^2/M^2])))] == 0$$

Sector 4, 0, coordinate 1, polynomial equation 1:

$$-216 + \text{KappaRefFinal}^4*\text{LambdaRefFinal} - 144*\text{Log}[\text{Mu}^2/M^2] + 2*\text{KappaRefFinal}^4*\text{LambdaRefFinal}*\text{Log}[\text{Mu}^2/M^2] == 0$$

Sector 4, 0, coordinate 2, polynomial equation 1:

$$216 - \text{KappaRefFinal}^4*\text{LambdaRefFinal} + 144*\text{Log}[\text{Mu}^2/M^2] - 2*\text{KappaRefFinal}^4*\text{LambdaRefFinal}*\text{Log}[\text{Mu}^2/M^2] == 0$$

Sector 4, 2, coordinate 1, polynomial equation 1:

$$-24 - \text{KappaRefFinal}^2*M^2 - \text{KappaRefFinal}^2*M^2*\text{Log}[\text{Mu}^2/M^2] == 0$$

Sector 4, 2, coordinate 2, polynomial equation 1:

$$24 + \text{KappaRefFinal}^2*M^2 + \text{KappaRefFinal}^2*M^2*\text{Log}[\text{Mu}^2/M^2] == 0$$

Sector 4, 2, coordinate 16, polynomial equation 1:

$$-72 - 3*\text{KappaRefFinal}^2*M^2 - 3*\text{KappaRefFinal}^2*M^2*\text{Log}[\text{Mu}^2/M^2] == 0$$

Sector 5, 0, coordinate 1, polynomial equation 1:

$$216*\text{Sqrt}[6]*\text{Kappa}*\text{Sqrt}[-(1/(M^2*\text{Kappa})^2*(1 + \text{Log}[\text{Mu}^2/M^2])))] + 144*\text{Sqrt}[6]*\text{Kappa}*\text{Sqrt}[-(1/(M^2*\text{Kappa})^2*(1 + \text{Log}[\text{Mu}^2/M^2])))] == 0$$

Sector 5, 0, coordinate 2, polynomial equation 1:

$$-(\text{KappaRefFinal}^5 * \text{LambdaRefFinal}) - 2 * \text{KappaRefFinal}^5 * \text{LambdaRefFinal} * \text{Log}[\sqrt{\mu}^2 / M^2] - \text{KappaRefFinal}^5 * \text{LambdaRefFinal}$$

Sector 5, 0, coordinate 4, polynomial equation 1:

$$\text{KappaRefFinal}^5 * \text{LambdaRefFinal} + 2 * \text{KappaRefFinal}^5 * \text{LambdaRefFinal} * \text{Log}[\sqrt{\mu}^2 / M^2] + \text{KappaRefFinal}^5 * \text{LambdaRefFinal}$$

Sector 5, 0, coordinate 5, polynomial equation 1:

$$3 * \text{KappaRefFinal}^5 * \text{LambdaRefFinal} + 6 * \text{KappaRefFinal}^5 * \text{LambdaRefFinal} * \text{Log}[\sqrt{\mu}^2 / M^2] + 3 * \text{KappaRefFinal}^5 * \text{LambdaRefFinal}$$

Sector 5, 0, coordinate 6, polynomial equation 1:

$$-7 * \text{KappaRefFinal}^5 * \text{LambdaRefFinal} - 14 * \text{KappaRefFinal}^5 * \text{LambdaRefFinal} * \text{Log}[\sqrt{\mu}^2 / M^2] - 7 * \text{KappaRefFinal}^5 * \text{LambdaRefFinal}$$

Sector 5, 0, coordinate 7, polynomial equation 1:

$$5 * \text{KappaRefFinal}^5 * \text{LambdaRefFinal} + 10 * \text{KappaRefFinal}^5 * \text{LambdaRefFinal} * \text{Log}[\sqrt{\mu}^2 / M^2] + 5 * \text{KappaRefFinal}^5 * \text{LambdaRefFinal}$$

Sector 6, 0, coordinate 1, polynomial equation 1:

$$-864 - 576 * \text{Log}[\sqrt{\mu}^2 / M^2] == 0$$

Sector 6, 0, coordinate 2, polynomial equation 1:

$$5184 + 3456 * \text{Log}[\sqrt{\mu}^2 / M^2] == 0$$

Sector 6, 0, coordinate 3, polynomial equation 1:

$$15552 + \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2 + 10368 * \text{Log}[\sqrt{\mu}^2 / M^2] + 3 * \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2$$

Sector 6, 0, coordinate 4, polynomial equation 1:

$$-36288 + \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2 - 24192 * \text{Log}[\sqrt{\mu}^2 / M^2] + 3 * \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2$$

Sector 6, 0, coordinate 5, polynomial equation 1:

$$-20736 + \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2 - 13824 * \text{Log}[\sqrt{\mu}^2 / M^2] + 3 * \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2$$

Sector 6, 0, coordinate 6, polynomial equation 1:

$$5184 - \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2 + 3456 * \text{Log}[\sqrt{\mu}^2 / M^2] - 3 * \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2$$

Sector 6, 0, coordinate 7, polynomial equation 1:

$$25920 - 5 * \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2 + 17280 * \text{Log}[\sqrt{\mu}^2 / M^2] - 15 * \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2$$

Sector 6, 0, coordinate 8, polynomial equation 1:

$$-8640 - \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2 - 5760 * \text{Log}[\sqrt{\mu}^2 / M^2] - 3 * \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2$$

Sector 6, 0, coordinate 9, polynomial equation 1:

$$-15552 - \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2 - 10368 * \text{Log}[\sqrt{\mu}^2 / M^2] - 3 * \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2$$

Sector 6, 0, coordinate 10, polynomial equation 1:

$$36288 + 17 * \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2 + 24192 * \text{Log}[\sqrt{\mu}^2 / M^2] + 51 * \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2$$

Sector 6, 0, coordinate 11, polynomial equation 1:

$$-160704 - 65 * \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2 - 107136 * \text{Log}[\sqrt{\mu}^2 / M^2] - 195 * \text{KappaRefFinal}^6 * \text{LambdaRefFinal} * M^2$$

8 Quadratic Four-Derivative Completion

The final exact-match statement requires the completed treatment of the leftover sector $\{2, 4\}$. The code path is `AnalyzeQuadraticFourDerivativeDifference`, which projects the verified remainder to the IBP basis, constructs a curvature-squared basis, constructs the quadratic higher-derivative redefinition image, and solves the resulting linear decomposition problem.

8.1 Verified $\{2, 4\}$ difference

$(9 \cdot \text{PD}[c] [\text{PD}[b] [h[a, -a]]] \cdot \text{PD}[-d] [\text{PD}[d] [h[-b, -c]]]) / (10 \cdot \text{Scalar}[M]^2) - (9 \cdot \text{PD}[-b] [\text{PD}[b] [h[a, -a]]] \cdot \text{PD}[-d] [\text{PD}[d] [h[-b, -c]]]) / (10 \cdot \text{Scalar}[M]^2)$

8.2 Projected $\{2, 4\}$ coordinates

$\{9/(10 \cdot \text{Scalar}[M]^2), -9/(20 \cdot \text{Scalar}[M]^2), -3/(5 \cdot \text{Scalar}[M]^2), 3/(10 \cdot \text{Scalar}[M]^2), -3/(20 \cdot \text{Scalar}[M]^2)\}$

8.3 Curvature-squared basis

Projected $R^{(1)2}$ expression:

$-2 \cdot \text{PD}[c] [\text{PD}[b] [h[a, -a]]] \cdot \text{PD}[-d] [\text{PD}[d] [h[-b, -c]]] + \text{PD}[-b] [\text{PD}[b] [h[a, -a]]] \cdot \text{PD}[-d] [\text{PD}[d] [h[c, -c]]]$

Projected $R^{(1)2}$ coordinates:

$\{-2, 1, 1, 0, 0\}$

Projected $R_{\mu\nu}^{(1)} R^{(1)\mu\nu}$ expression:

$-1/2 \cdot (\text{PD}[c] [\text{PD}[b] [h[a, -a]]] \cdot \text{PD}[-d] [\text{PD}[d] [h[-b, -c]]]) + (\text{PD}[-b] [\text{PD}[b] [h[a, -a]]] \cdot \text{PD}[-d] [\text{PD}[d] [h[c, -c]]])$

Projected $R_{\mu\nu}^{(1)} R^{(1)\mu\nu}$ coordinates:

$\{-1/2, 1/4, 1/2, -1/2, 1/4\}$

8.4 Quadratic higher-derivative field redefinition image

Independent rank-2 source basis elements selected by the code:

$\{\eta[u, v] \cdot \text{PD}[-z\$32101] [\text{PD}[-z\$32100] [h[z\$32100, z\$32101]]], \text{PD}[-z\$32143] [\text{PD}[z\$32143] [h[u, v]]]\}$

Independent projected image vectors:

$\{-2, 0, 2, 0, 0\}, \{2, -1, 0, -2, 1\}, \{2, -2, 0, 0, 0\}$

8.5 Solved decomposition

Linear solve output:

$\langle | \text{"HasSolution"} \rightarrow \text{True}, \text{"Variables"} \rightarrow \{c241, c242, c243, c244, c245\}, \text{"Solution"} \rightarrow \{\{c243 \rightarrow -\dots\}\} \rangle$

Curvature coefficients:

$\langle | \text{"R2"} \rightarrow 0, \text{"Ricci2"} \rightarrow 0 | \rangle$

Quadratic redefinition coefficients:

```
<|1 -> -3/(10*Scalar[M]^2), 2 -> -3/(20*Scalar[M]^2), 3 -> 3/(10*Scalar[M]^2)|>
```

Reconstructed quadratic redefinition expression:

```
(9*PD[c][PD[b][h[a, -a]]]*PD[-d][PD[d][h[-b, -c]]])/(10*Scalar[M]^2) - (9*PD[-b][PD[b][h[a, -a]]])/(10*Scalar[M]^2)
```

Residual expression:

```
0
```

Residual zero check: True. Pure quadratic redefinition check: True.

9 Independent Cross-Check

A second script reproduced the $\{2, 4\}$ statement directly from the saved vectors, without calling the main analyzer helper.

9.1 Target coordinates

```
{9/(10*Scalar[M]^2), -9/(20*Scalar[M]^2), -3/(5*Scalar[M]^2), 3/(10*Scalar[M]^2), -3/(20*Scalar[M]^2)}
```

9.2 Independent redefinition rows and vectors

```
{3, 4, 5}
```

```
{{-2, 0, 2, 0, 0}, {2, -1, 0, -2, 1}, {2, -2, 0, 0, 0}}
```

9.3 Pure redefinition solve

```
{{p[1] -> -3/(10*Scalar[M]^2), p[2] -> -3/(20*Scalar[M]^2), p[3] -> 3/(10*Scalar[M]^2)}}
```

```
{p[1] -> -3/(10*Scalar[M]^2), p[2] -> -3/(20*Scalar[M]^2), p[3] -> 3/(10*Scalar[M]^2)}
```

```
{9/(10*Scalar[M]^2), -9/(20*Scalar[M]^2), -3/(5*Scalar[M]^2), 3/(10*Scalar[M]^2), -3/(20*Scalar[M]^2)}
```

9.4 Curvature basis vectors

```
<|"R2" -> {-2, 1, 1, 0, 0}, "Ricci2" -> {-1/2, 1/4, 1/2, -1/2, 1/4}|>
```

9.5 Joint curvature plus redefinition solve

```
{{q[1] -> -1/2*cR2 - cRicci2/4 - 3/(10*Scalar[M]^2), q[2] -> -1/4*cRicci2 - 3/(20*Scalar[M]^2)}
```

```
{q[1] -> -1/2*cR2 - cRicci2/4 - 3/(10*Scalar[M]^2), q[2] -> -1/4*cRicci2 - 3/(20*Scalar[M]^2)}
```

```
{9/(10*Scalar[M]^2), -9/(20*Scalar[M]^2), -3/(5*Scalar[M]^2), 3/(10*Scalar[M]^2), -3/(20*Scalar[M]^2)}
```

9.6 Cross-check summary

```
<|"PureRedefinitionHasSolutionQ" -> True, "PureRedefinitionReconstructionMatchesQ" -> True, "JointCurvaturePlusRedefinitionMatchesQ" -> True|>
```


10 Final Verdict

The stored final result satisfies:

```
<|"ExactMatchQ" -> True, "UnsupportedSectors" -> {}, "VerifiedSupportedMatchQ" -> True, "Quadr
```

Therefore, using the exact stored branch and the exact implemented quotient operations, the mass-dimension-six EFT is equivalent to the sign-flipped Einstein–Hilbert reference.